

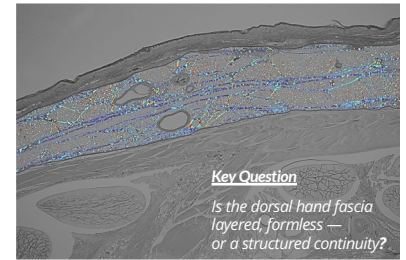
Structured continuity of the dorsal hand fascia

quantitative directional decomposition of collagen orientation

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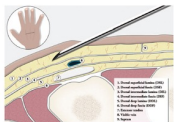


ANGLE TO LOCAL SKIN TANGENT

- Tangential < 30°
- Oblique 30–60°
- Perpendicular > 60°

BACKGROUND & GAP

Rejuvenation and reconstruction of the dorsal hand depend on fascial planes that remain contested. Its fascia has been described either as three discrete, stacked layers (Bidle 2010) or as a continuous, sponge-like fibrous web with no true layers (Lefebvre-Vilardebo 2015). Both accounts are qualitative — the directional organization of the collagen that would reconcile them has never been quantified. A recent qualitative dissection study independently describes this continuity — dense oblique fascial-dermal attachments through the web space, and dorsal skin loosely tethered proximally but firmly held distally (Dowlatshahi et al. 2026, Anat Sci Int). What has not been done is to measure it.



A Layered Architecture of Dorsal Hand, with injection cannula placed within the dermal lamina, (Bidle SM et al., 2010)

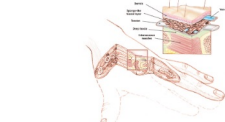


Illustration of the 3D spongelike fascial framework on the dorsal side of hands: (Lefebvre-Vilardebo et al., 2015)

AIM

Quantify collagen orientation with a structure-tensor pipeline to reconcile these models — testing whether the dorsal hand fascia is instead a **structured continuity** whose tangential-sheet and perpendicular-bridge composition reorganizes, zone by zone, along the hand.

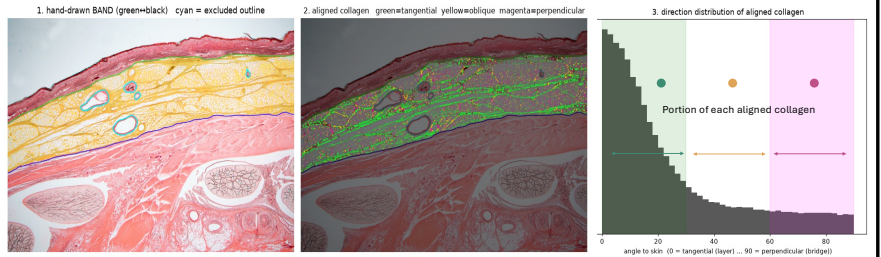
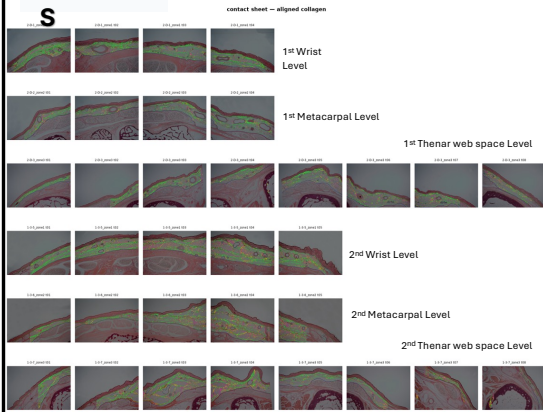
CLINICAL RELEVANCE

By mapping where the dorsal skin is anchored versus where it glides, these directional maps may offer an anatomical basis for selecting injection planes in dorsal-hand rejuvenation — **helping avoid the bridge-dense zones where filler may tether or distort mobile skin**. It also frames skin glide as a rehabilitation target — where edema or scarring abolish the tangential plane and stiffen finger motion. A hypothesis for validation, not yet a guideline.

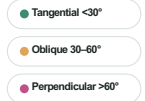


Before and after filler injection on dorsal hand 2018, JS Kim

METHOD



- Two cadaveric hands; undecalcified histology (EXAKT, H&E) at three transverse levels: wrist, metacarpus, thenar web space.
- Band ROI: skin → deep fascia roofing the extensor tendons; tendons excluded.
- Structure tensor (scikit-image) classifies aligned collagen by its angle to the local skin tangent:
- Open-source pipeline: automated fiber analysis with a manually delineated band ROI (Skin-tangent reference).
- 34 fields analyzed; the specimen taken as the unit of inference (n = 2).



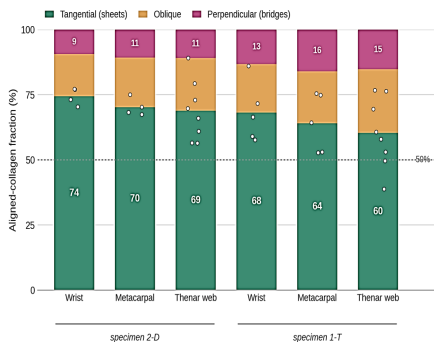
RESULTS

Composition

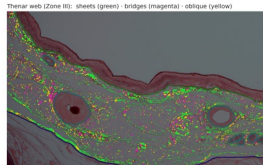
Sheets dominate everywhere — bridges are always present.

Tangential sheets dominate every zone; bridges are always present

Section means across two cadaveric hands; white dots = individual fields



Tangential collagen predominates in all 34 fields (section means 60–74%). Perpendicular bridging fibers are present in every field (section means 9–16%, up to 28% locally) — coexistence, not either model alone.

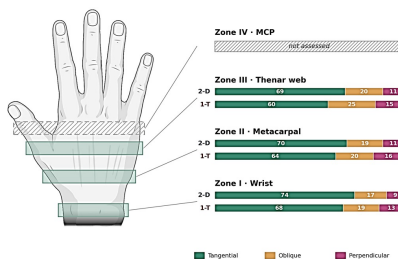


Level gradient

Level by level: sheets fall, oblique rises, bridges persist.

Directional composition along the dorsal hand

Bars = the transverse section analysed at each level (n = 2 hands, per specimen), not the whole zone. Both hands independently show sheets fall and oblique rises, proximal → distal. Zone IV (MCP) not assessed.

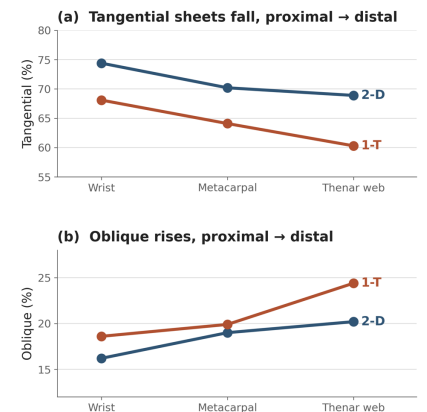


Proximal → distal: tangential 71 → 65%, oblique 17 → 22%; perpendicular present at every level. Both hands independently show the same proximal-to-distal decline; within the thenar web perpendicular rises toward the margin (preliminary — pending band-boundary standardization). The gradient replicates across specimens. The same proximal-to-distal shift has been described qualitatively from cadaveric dissection (Dowlatshahi et al. 2026, Anat Sci Int), independent of the present measurements.

Replication

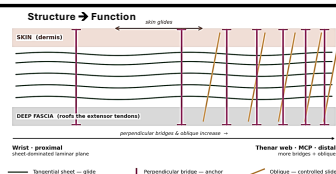
Both hands, same direction — the gradient replicates across specimens (n = 2)

Each hand analysed independently; specimen means per level.



Interpretation

Perpendicular bridges act as reticular-cutis-like tethers (anchoring), while oblique septa permit controlled skin sliding (glide). Both rise toward the mobile web / MCP — these directional maps trace where dorsal skin must glide and stay anchored.



Key Message

The dorsal hand fascia is neither layered nor formless, but a structured continuity — persistent tangential sheets interwoven with perpendicular bridges. Bridges concentrate where the skin must glide and stay anchored, forming a proximal-to-distal mobility gradient.

REFERENCES: Bidle 2010 · Lefebvre-Vilardebo 2015 · Dowlatshahi et al. 2026, Anat Sci Int. The study was supported by an educational grant from Merz Aesthetics (Grant No. 2025-31-1015).